

Parallel AC Circuits

①

Parallel circuits are those in which the various circuit elements are connected in parallel to each other. In such circuits, the voltage is same across each branch.

Some features of parallel circuits

- ⇒ current in any branch ~~is~~ depends upon the impedance of that branch.
 - ⇒ The total line current supplied to the circuit is the phasor sum of branch currents.
 - ⇒ In pure parallel circuit, each element ~~is~~ connected shares the same voltage.
 - ⇒ Electrical systems uses parallel connections more than series.
- for example : → Electrical devices and equipment are connected in parallel across a.c mains.

Types :-

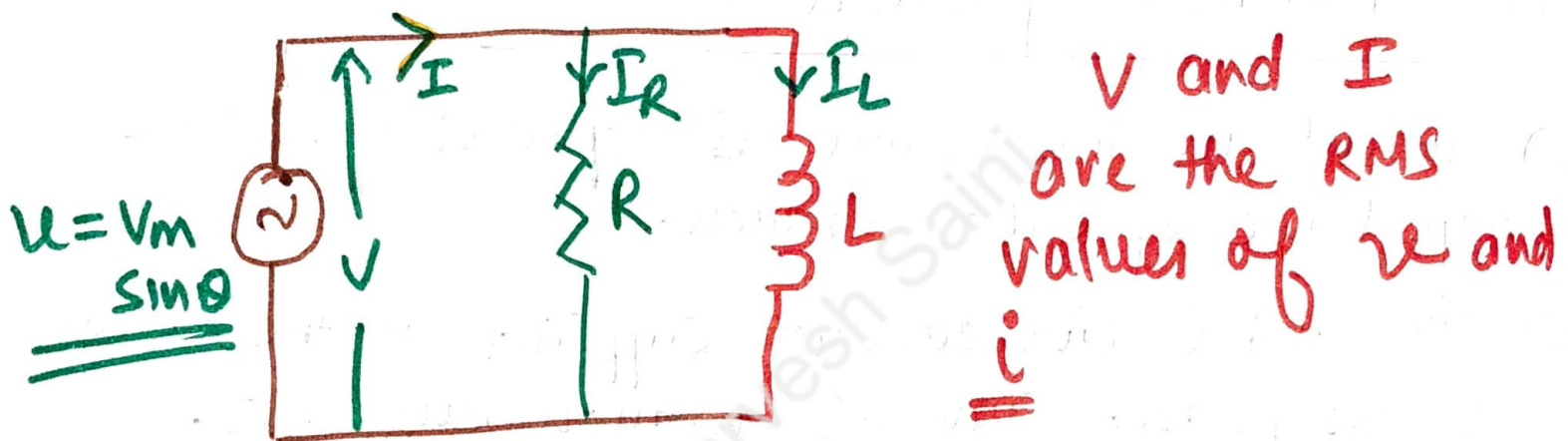
- Parallel R-L circuit
- Parallel R-C circuit
- Parallel RLC "

Parallel R-L AC Circuits

(2)

In R-L parallel ac circuit, a resistor (R) and an Inductor (L) are connected in parallel with each other across an ac voltage source.

The circuit diagram is shown below:



The instantaneous voltage is given by:

$$u = V_m \sin \theta \quad \text{--- ①}$$

In this circuit, the current (I) lags the voltage (V) by angle " ϕ ".
Because it is an inductive circuit.

I_R and I_L are the RMS values of current through "R" and "L". (3)

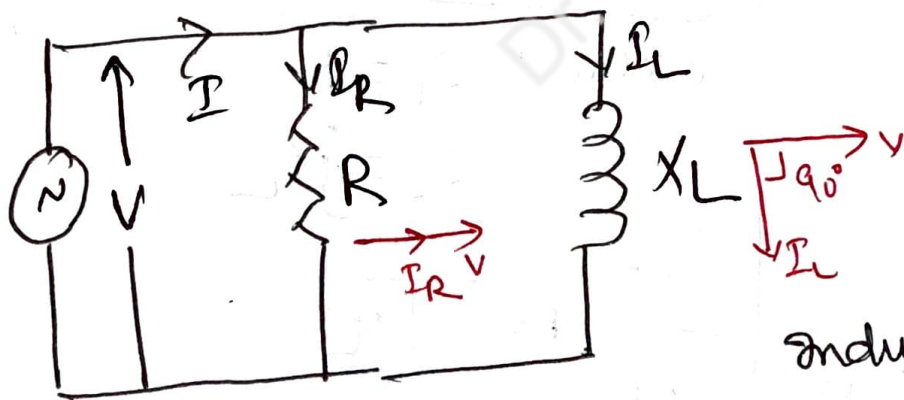
The Inductive reactance is given by:

$$X_L = 2\pi fL$$

where, $f \Rightarrow$ frequency of the input signal (i.e. ω).

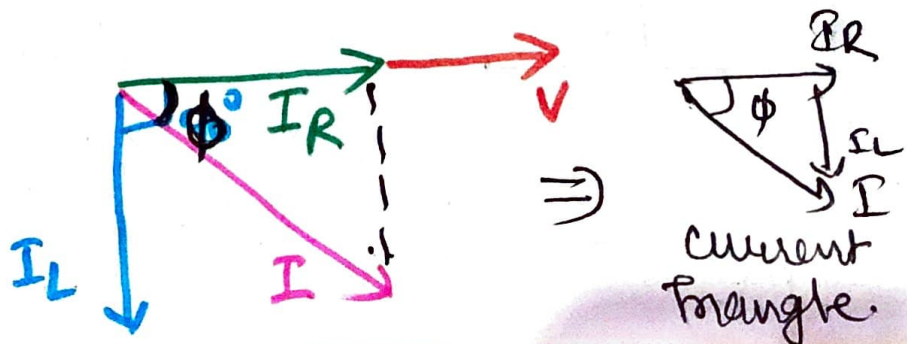
Phasor diagram : Now let us draw the phasor diagram of parallel R-L circuit.

Since, voltage is the common factor; hence the voltage phasor (V) is taken as the reference phasor. The diagram is shown below: $\rightarrow$



as we know that the phase difference ~~between~~ between V & I_R in ~~resistor~~ Resistor is ZERO. and on inductor, I_L lags V by 90° .

The resultant phasor diagram of this circuit is:-



Analysis of parallel RL ac circuit

(4)

The total current (I) in the circuit can be determined as:

$$I = \frac{V}{Z} \quad \text{--- (1)}$$

where, $Z =$ Impedance of the circuit.

In parallel circuits, instead of impedance (Z), we prefer Admittance (Y)

where, admittance is the reciprocal of Z and is measured in Siemens (S) [or mho Ω^{-1}]

Similarly, instead of Resistance, we prefer conductance (G). It is given by

$$G = \frac{1}{R}$$

and, instead of reactance we use susceptance (B).

Susceptance (B) is given by $B = \frac{1}{X}$ $X \Rightarrow$ Reactance

G and B are also measured in Siemens (S)

Hence the impedance (Z) of the given parallel RL ac circuit is

$$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2} \quad \text{--- (2)}$$

$$\text{where, } Z_1 = R \\ Z_2 = jX_L$$

$$\frac{1}{Z} = \frac{1}{R} + \frac{1}{jX_L}$$

or

$$Y = G + \frac{B_L}{j} \times \frac{j}{j} = G - jB_L$$

$$Y = G - jB_L \quad \text{--- (3)}$$

where, $B_L \Rightarrow$ Inductive susceptance.
UNIT of $B_L =$ Siemens (S)

The ~~admittance~~ admittance approach is very useful in the solution of parallel a-c circuits.

Now, coming back to the analysis.

$$I = \frac{V}{Z} \quad \text{or} \quad \text{~~admittance~~}$$

or

$$I = V \times Y \quad (\because \frac{1}{Z} = Y)$$

$$I = V(G - jB_L)$$

$$\left(\frac{I}{V}\right) = G - jB_L$$

$$Y = G - jB_L \quad \leftarrow \text{This is same as eqn (3)}$$

The total current I is

$$I = I_L + I_R$$

$$I_R = \frac{V}{R}$$

$$I_L = \frac{V}{jX_L}$$

$$I = \frac{V}{Z}$$

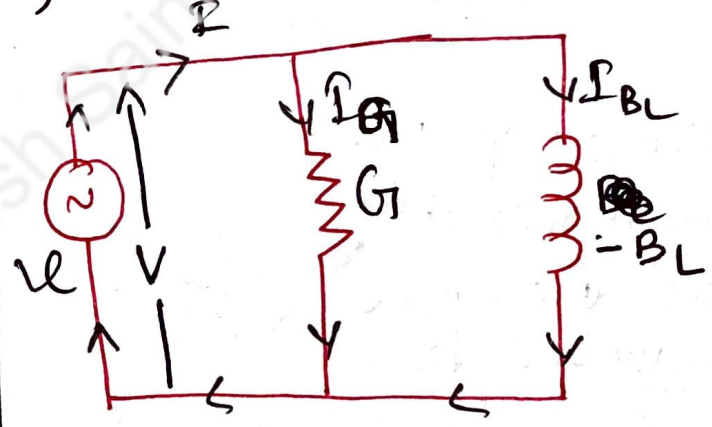
hence,

$$\frac{V}{Z} = \frac{V}{R} + \frac{V}{jX_L}$$

$$\frac{1}{Z} = \frac{1}{R} + \frac{1}{jX_L}$$

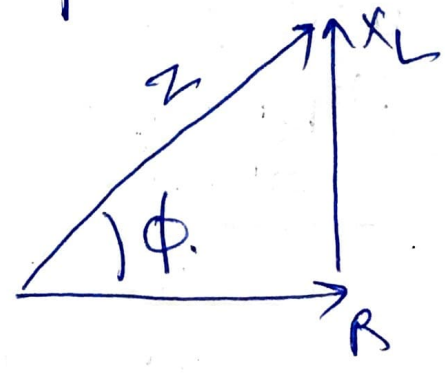
$$Y = G - jB_L \quad \leftarrow \text{again same as eqn (3)}$$

The given circuit diagram of parallel RL circuit can be drawn in terms of Y , G and B_L as:



The impedance and ~~admittance~~ admittance Δ s are:

(1) Impedance Δ .



1) Admittance Δ

6

Since, G and B_L are the components of $\underline{Y}$;
 So, $\underline{G}$ is the component, which is in-phase with the reference phasor (i.e V) and B_L is the one which lags reference phasor (i.e V) by 90° . This B_L component is also known as Quadrature component of Y .

The admittance Δ is a right angle Δ which is shown below:

Hence; Y can be represented as a complex number as:

$$Y = G - jB_L$$

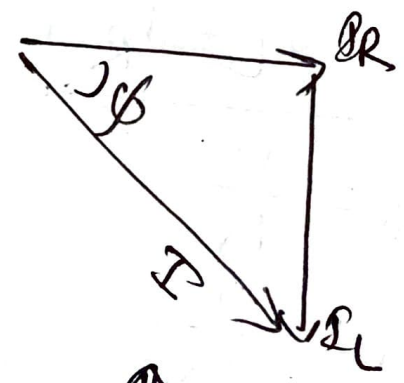
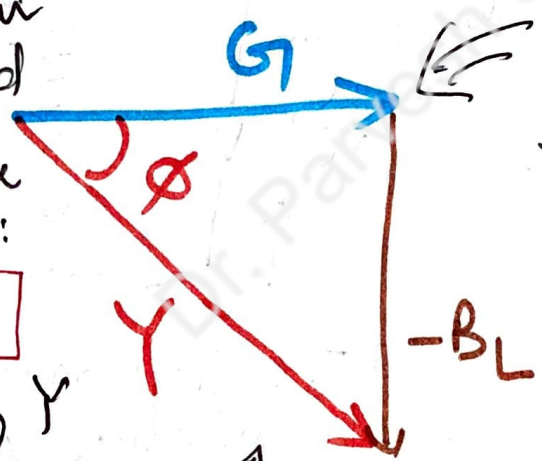
magnitude of Y

$$|Y| = \sqrt{G^2 + B_L^2}$$

Phase angle ϕ

$$\phi = \tan^{-1} \frac{B}{G}$$

admittance Δ is obtained from voltage current Δ . as shown below:

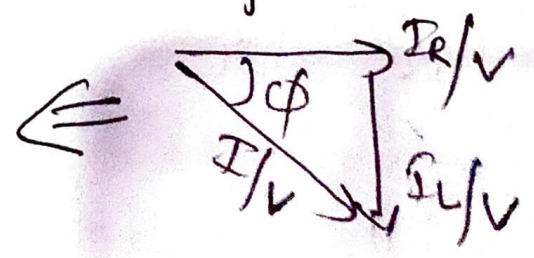


divide each phasor with reference volt (V) ~~reference voltage~~ ~~reference voltage~~ we get

Now;

$$\frac{R}{V} = \frac{1}{Z} = Y \quad \frac{X_L}{V} = \frac{1}{Z} = Y$$

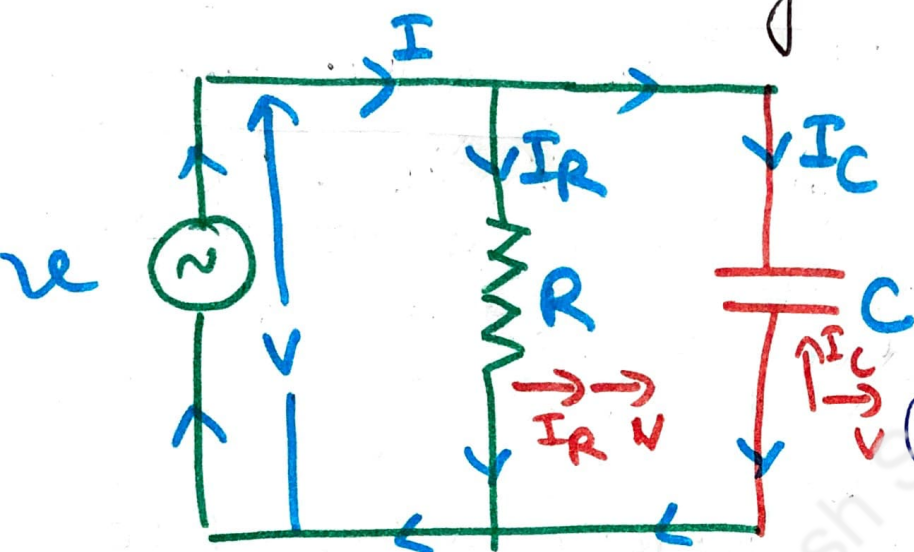
$$\frac{R}{V} = \frac{1}{Z} = Y$$



Parallel R-C a.c. Circuit :-

(7)

A parallel R-C ac circuit is the one in which Resistor (R) and Capacitor (C) are connected in parallel with each other across an ac voltage source. The circuit diagram is shown below:



where, V and I are the RMS values of the instantaneous voltage (v) and current (i).

The instantaneous value of voltage is given by:

$$v = V_m \sin \theta$$

— (1)

Now, because it is a capacitive circuit. Hence, the current (I) leads the voltage (V) by an angle " ϕ ". Hence the instantaneous current is given by:

$$i = I_m \sin (\theta + \phi)$$

— (2)

The total current, I is given by:

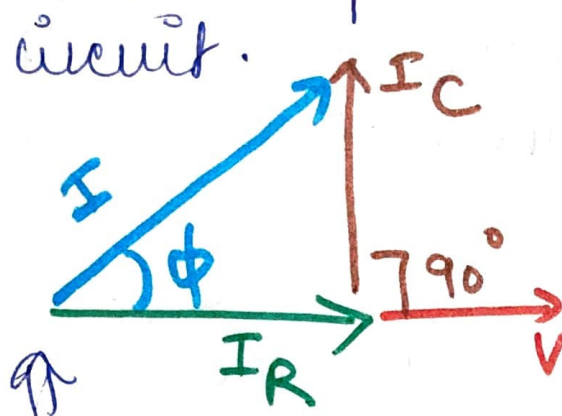
$$I = I_R + I_C$$

— (3)

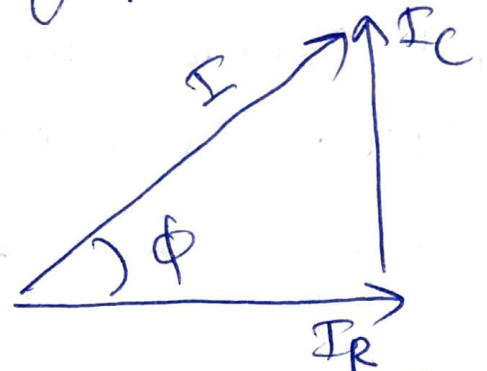
where, I_R and I_C are the RMS currents through resistor and capacitor respectively.

Analysis of parallel R-C ac circuit: (8)

Taking voltage (V) as the reference phasor, we can draw the phasor diagram of parallel R-C ac circuit.



This is the phasor diagram of given parallel RC ac circuit.



This is current Δ .

From eqⁿ (3) we have

$$I = I_R + I_C$$

$$I = \frac{V}{Z} ; I_R = \frac{V}{R} \text{ and}$$

$$I_C = j \frac{V}{X_C}$$

So; Substitute in eqⁿ (3)

$$\frac{V}{Z} = \frac{V}{R} + j \frac{V}{X_C}$$

or,

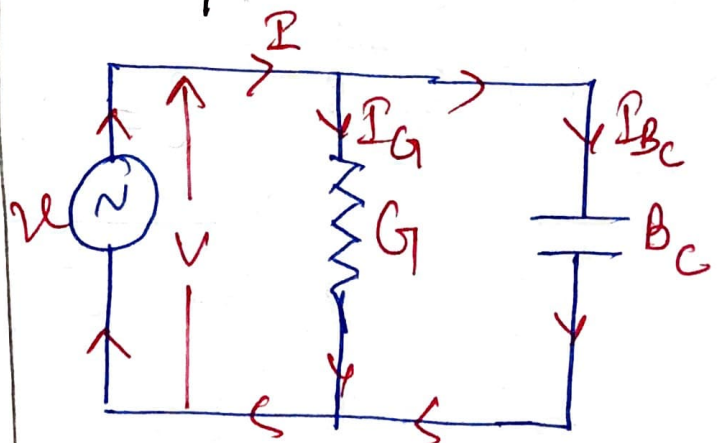
$$\frac{1}{Z} = \frac{1}{R} + j \frac{1}{X_C}$$

or

$$\boxed{Y = G + j B_C} \quad \text{--- (4)}$$

where, $B_C \Rightarrow$ capacitive susceptance (measured in Siemens, S).

The given circuit diagram in terms of G and B_C can be drawn as:

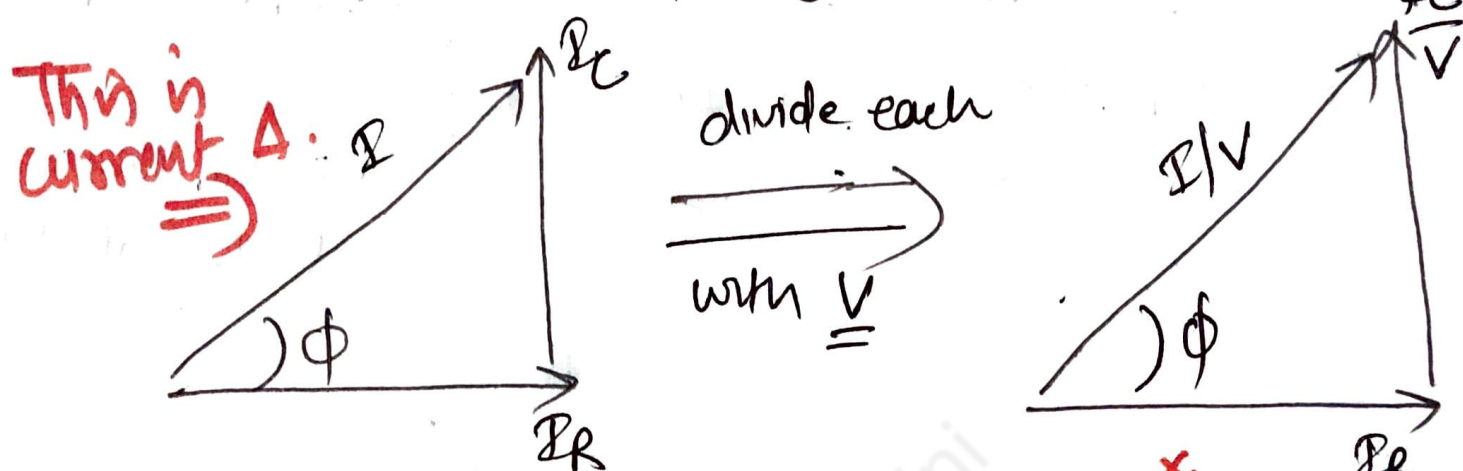


G and B_C are the components of Y . (as given in eqⁿ 4)

Impedance and Admittance Triangle

9

The admittance triangle can ~~be~~ be obtained from the current triangle as shown below:



as we know:

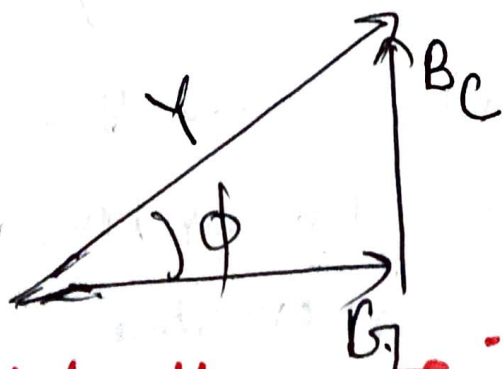
$$\left[\frac{I}{V} = \frac{1}{Z} = Y \right]$$

$$\left[\frac{I_C}{V} = \frac{1}{X_C} = B_C \right]$$

$$\left[\frac{I_R}{V} = \frac{1}{R} = G \right]$$

This is Admittance Δ

hence, the given current Δ can be converted into Admittance Δ as shown below:



Admittance Triangle

$$Y = G + j B_C$$

$$|Y| = \sqrt{G^2 + B_C^2}$$

phase angle, ϕ is

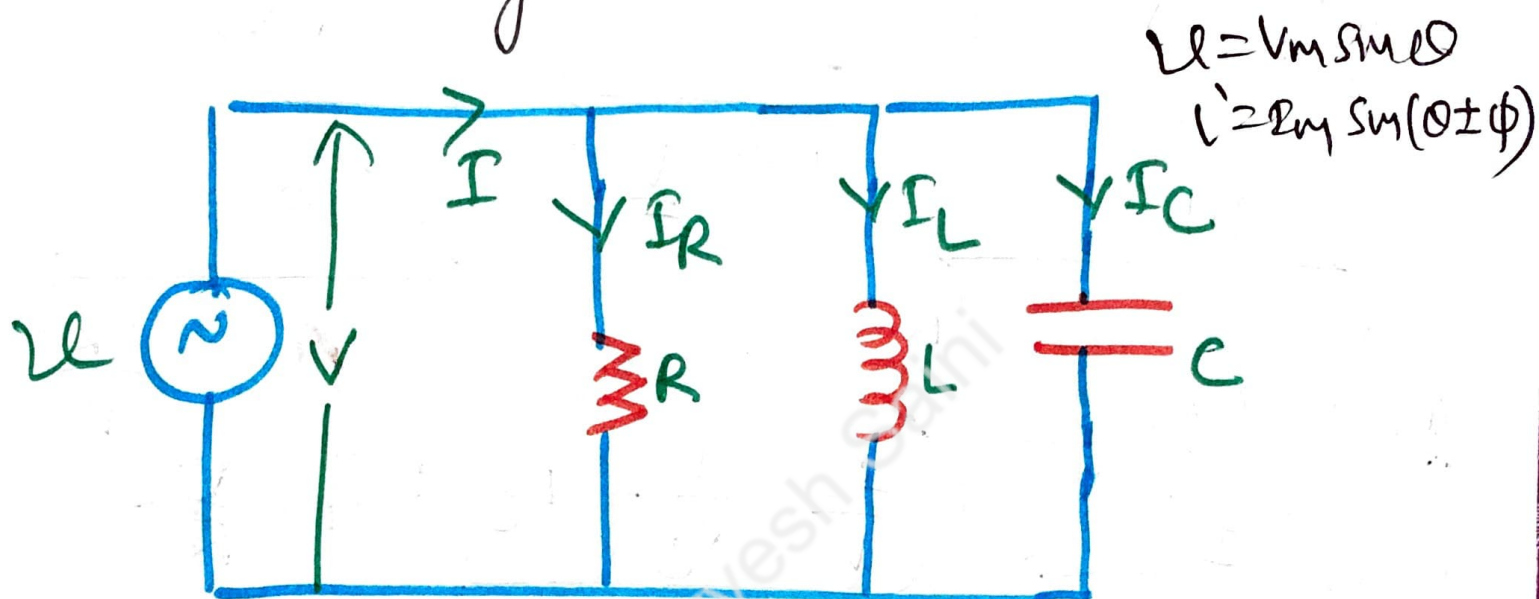
$$\phi = \tan^{-1} \frac{B_C}{G}$$

Parallel RLC ac circuit:

(10)

Let us consider an ac circuit in which all the elements R , L and C are connected in parallel with each other across an ac voltage source.

The circuit diagram is shown below:



I_R , I_L and I_C are the RMS currents through R , L and C respectively.

I and V are the RMS current and voltage in the given ac circuit.

$$I = I_R + I_L + I_C \quad \text{--- (1)}$$

$$I = \frac{V}{Z} = V Y \quad (\because \frac{1}{Z} = Y) \Leftarrow \text{admittance}$$

$$I_R = \frac{V}{R} = V G \quad (\because \frac{1}{R} = G) \Leftarrow \text{conductance}$$

$$I_C = \frac{V}{X_C} = V B_C \quad (\because \frac{1}{X_C} = B_C) \Leftarrow \text{capacitive and}$$

$$I_L = \frac{V}{X_L} = V B_L \quad (\because \frac{1}{X_L} = B_L) \Leftarrow \text{inductive susceptance.}$$

Substituting all in eqⁿ ① we get

$$\frac{V}{Z} = \frac{V}{R} - j \frac{V}{X_L} + j \frac{V}{X_C}$$

$$\text{or, } \frac{1}{Z} = \frac{1}{R} - j \frac{1}{X_L} + j \frac{1}{X_C}$$

$$\frac{1}{Z} = \frac{1}{R} + j \left(\frac{1}{X_C} - \frac{1}{X_L} \right)$$

$$Y = G + j(B_L - B_C) \quad \text{--- ②}$$

Now, from eqⁿ ② we can see that there is net effect of susceptance in the circuit.

This $B_L - B_C > 0$ (i.e. +ve) $\Leftrightarrow$ Case I

$B_L - B_C < 0$ (i.e. -ve) $\Leftrightarrow$ Case II

$B_L - B_C = 0$ | Special case. ~~Section~~

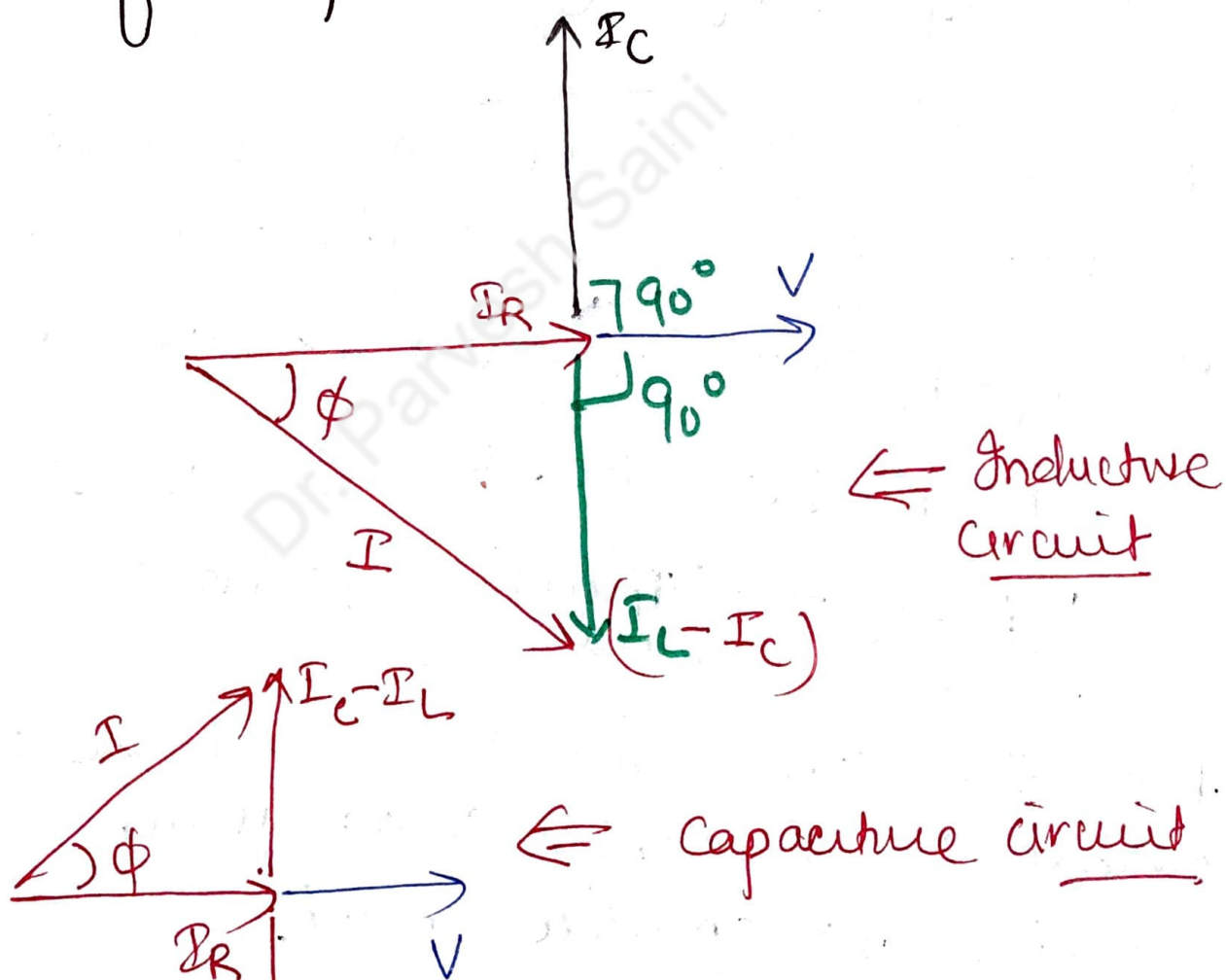
From the Case I and II it can be observed whether the given RLC circuit is an inductive or capacitive circuit.

If the given RLC parallel ac circuit is an inductive, then the phase angle is negative and hence the current (I) lags the voltage (V) by an angle ϕ .

Similarly, If the given ~~AC~~ RLC parallel ac circuit is a capacitive circuit, then, the Current (I) leads the voltage (V) by angle ϕ° . (12)

Analysis $\rightarrow$

1) Phasor diagram :- Taking voltage (V) phasor as reference, we can draw the phasor diagram of a parallel RLC ac circuit.



From eqn (3) magnitude of Y is calculated as:

$$|Y| = \sqrt{G^2 + (B_L - B_C)^2}$$

$\rightarrow$ (4)

and the phase ϕ is

$$\phi = \tan^{-1} \frac{B_L - B_C}{G} \rightarrow (5)$$

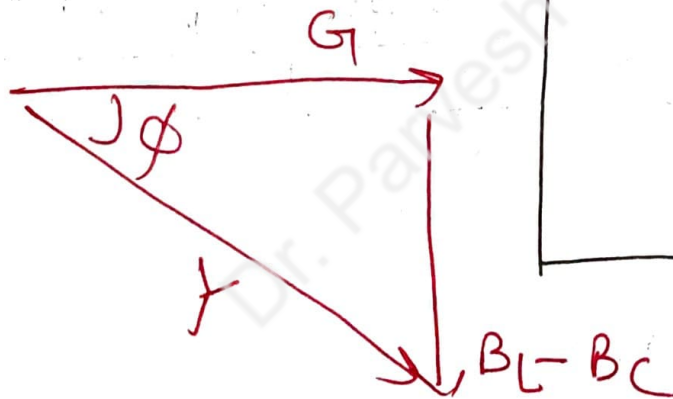
14 eqⁿ (5); If $B_L > B_C$ then the phase angle is +ve, and
 If $B_L < B_C$, then the phase angle ϕ is -ve

Impedance Δ / or Admittance Δ .

The admittance Δ for a parallel RLC ac circuit (for both cases) are shown below:

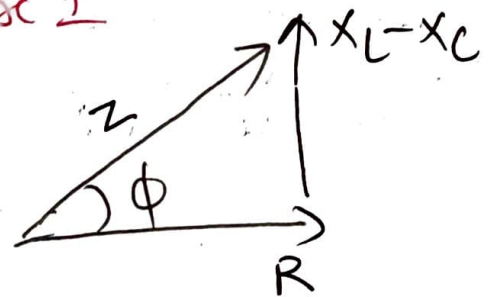
For inductive case :-

Case I



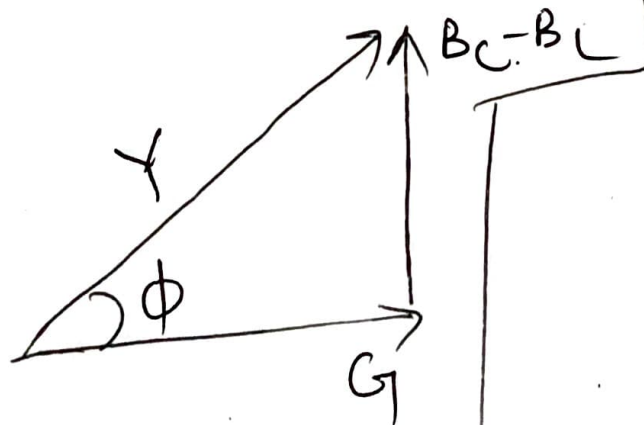
Impedance Δ

Case I



For capacitive case :-

Case II



Case II

